# Sony MA900 Start Up/Shutdown: Sacramento

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Note: Always handle the instrument with aseptic techniques. EtOH your gloves and tanks when you are handling the instrument. Additionally, when working with your samples in the bio-bubble, be mindful to clean the table and placement of covers.

1) Always handle the system and computer with gloves on.
2) The computer should be on.
    a. If it is not: the power button for the computer is on the monitor itself found on the bottom right. The Windows login information:
        i. Username login: fcm
        ii. Password: fcm
            i. Note: If you run into any issues reference the Troubleshooting Step 1
3) System fluids maintenance and setup:
    a. Open the Biobubble access flap, spray 70% ETOH on the working surface in front of the MA900.
    b. Check system fluids:
        i. Press the wetcart brake release button with your shoe.
        ii. Slowly, pull the wetcart slowly out from under the instrument table until it is clear of the table top.
        iii. Press the wetcart brake button with your shoe to lock it in place.
        iv. Prepare Sheath Tank:
            i. Set the plastic holder tray (Located on the wetcart on its side) to the left side of the sheath tank.
            ii. ETOH top of the tank and your gloves.
            iii. Disconnect the quick disconnect for the air line (the clear tubing line) and set into the holder tray.
            iv. Disconnect quick disconnect for the main sheath line (blue tubing line) and set into the holder tray.
            v. Carefully remove the sheath tank lid and set upside down inside the cleared working surface in the Biobubble.
            vi. Lift the sheath tank carefully out of the fluidics cart and place it onto the step stool under the sheath cube to the left of the computer desk.
            vii. Position the sheath cube spout over the sheath tank opening (move the tank as close as possible then reposition the sheath cube as needed) and fill the sheath tank to the **weld line**.

viii. Return the tank to the wetcart and reinstall the lid, airline and sheath line.

v. Prepare the waste tank:
    i. Disconnect the waste line using the quick disconnect from the top of the waste tank.
    ii. Lift the tank and move it to the sink counter.
    iii. Place a large Kimwipe on the counter next to the waste tank.
    iv. Carefully remove waste lid and put it upside down on Kimwipe.
    v. If liquid is preset in the tank, dump down the sink.
    vi. Add 1L of fresh 100% bleach to the waste tank.
    vii. Replace the waste tank lid
    viii. Return the waste tank to the wet cart
    ix. Replace the waste tubing.
vi. Press the wetcart brake release button with your shoe.
vii. Slowly, push the wetcart slowly back under the instrument table.
viii. Press the wetcart brake button with your shoe to lock it in place.

4) Power on the Air compressor by flipping the toggle switch found on the back of the compressor on the right side. To check if it is on you will see the green light turn on in the front of the air compressor.
5) Power on the MA900 instrument by pressing the Power/Standby button
    a. While the instrument is powering on. Wipe down/de-static the sort chamber surfaces.
        i. Clean the deflection plates and side-stream windows wipe with a kimwipe that has some EtOH.
        ii. To remove the deflection plates, unscrew the light brown pegs attached to the plate holder. Turn counter-wise to loosen and clockwise to tighten.
    b. Remove any collection holder from the sort chamber

Note: **Do not turn on the software** until the instrument display is on STANDBY, if you try to turn on the software before the instrument says STANDBY this can cause connection issues.

6) Open the Sony MA900 software: (**IMPORTANT**: Note the two methods: new nozzles vs used nozzles)
    *Note: Please **reuse** your nozzles for up to 1 month from initial use date.
    a. For **new** Nozzles:
        i. Open the software found on the desktop called MA900.
        ii. Obtain the foil nozzle package from the shelf unit for the correct size.
        iii. IMPORTANT: Label the nozzle package with the following:
            i. Lab Name
            ii. Date
            iii. Time
    b. For **used** nozzles:
        i. Open the “RunAsDate” folder on the desktop
        ii. Run the “RunAsDate” program.

iii. Set the date and time to the date and time on the used nozzle foil package.
iv. Click “Run”

c. Log into software:
    i. Username Login/Password is unique for each lab, login details can be found in the User Account text document on the desktop.

7) After you are logged in follow the on-screen wizards called Initial Instrument Setup
    a. On Step 1 Chip detection, grab the nozzle chip size you want to use (70um,100um,130um) and scan it by showing it to the computer camera.
        i. Nozzle chips are good for use within 24 hours. Check the most recent wrapper if you are using the same nozzle size. Can reuse the chip if within 24 hours of last use.
        ii. If unable to find the computer camara. Can be found on the center top of the monitor. Press down the top center and it should pop up.
        iii. Scan the nozzle wrapper denote the date and time on the wrapper.
    b. On Step 3 Chip loading step. Open the flip-up door button

    Note: If you run into issues refer to the Sony Troubleshooting Step 2 section b.
        i. If reusing the chip reinsert the chip, by pressing it back down into the instrument until it recognizes the chip. Once it recognizes the chip the instrument will automatically take over in the placement of the chip.
        ii. Using a new chip remove the current sorting chip and insert the new chip. When inserting the chip into the slot have the Sony logo and nozzle size mark at the top facing toward you and reading the information from left to right.
    c. On Step 4 Laser setting, be sure to check mark all the laser to have them all warmed up and on.

    Note: If you run into issues refer to the Sony Troubleshooting Step 2 section c.

        i. The instrument lasers do take about 10 minutes for it to warm up
        ii. Once it is warmed up you will be able to proceed to the next step

    d. On Step 5 Filter setting you will be shown 2 options. Select the standard option for the instrument set-up
        i. We do not have the filters for the fluorescence protein setting.

    e. On Step 6 Fluidics check

    Note: If you run into issues refer to Sony Troubleshooting Step 2 section e.

i. Sheath fluid droplets will appear from the tip of the sample probe and on the window screen. If not, the sample line could be clog. When this step is complete the image droplet should look still as a picture. If you notice that shaking, you will need to troubleshoot this.

8) Running Automatic Calibration follow the on-screen instructions (Contains a status bar per step)

Note: If the status bar refreshes 3x times over on any of the steps or error message appears. Reference the Sony Troubleshooting Steps or grab/email a core member.

a. Prepare the Automatic Setup Beads
    i. Found in the cold room next main core lab (insert room#3423)
    ii. Check if there is a prepare tube. The beads do settle at the bottom of the tube. Resuspend the bead sample. The solution should turn blue.
    iii. If you need to make a new tube of beads. Vortex or shake the bottle vigorously and add in approximately 15 drops (1mL) of undiluted beads.
    iv. Place the correct sample tube holder into the sample chamber. To install the tube holder, align the D-shaped notch on the bottom of the sample tube holder with the slot in the top of the agitation unit
        i. Note: Do not place any caps in the sample holder it will cause the instrument to not pressurize the sample chamber accordingly and will not run the sample.
    v. Once all loaded Start the Auto-Calibration sequence

b. On Step 1 Chip alignment and Laser Delay Setting make sure to have both checked mark on when you are sorting

Note: If the status bar refreshes 3x times over on this step or error message appears. Reference Sony Troubleshooting Step 3 Section b

    i. If you just want to use analysis only, you can select skip auto calibration
    ii. If you are using the 100 micron the onboard screen will prompt you to either select Standard or targeted cells (Large Cells)
        i. Select Standard

c. On Step 2 Droplet Calibration

Note: If the status bar refreshes 3x times over on this step or error message appears. Reference Sony Troubleshooting Step 3 Section c

    i. Check the stability of the stream. If not stable reference the troubleshooting steps and follow the on-screen troubleshooting wizards steps

d. On Step 3 Side Stream Calibration

Note: If the status bar refreshes 3x times over on this step or error message appears. Reference Sony Troubleshooting Step 3 Section d.

i. Check if the sort chamber is empty and does not have a collection holder inside
ii. The deflection plates could have salt build up and will need to be recleaned with a kimwipe and EtOH.
iii. Clean the sort chamber walls and base with a kimwipe and EtOH

e. On Step 4 Sort Delay Calibration
    i. Check if you ran out of beads

9) Create a New Experiment
    a. Find your template or create a new experiment
    b. Tips when running
        i. When you hit start and want to analyze the sample and not having it continuously running. Hit the pause button to avoid sample loss. The system will automatically backflush when you hit the stop button.
        ii. When you hit record or start sort on a tube. Then you stop that tube you are unable to reselect or continue running with that tube. You will need to create a new tube if you want to continue running the same sample.
            i. The sort data is automatically saved in the tube, you can reference the sort numbers by selecting on the tube. In the toolbar ribbon click tube result->then open the tab label sorting results.
        iii. If you need to add dilute buffer or resuspend the sample. Hit pause and hit the unloaded sample button found under the Start/Pause button

10) How to Sort
    a. Once you have set your target population with the appropriate gating strategy, you can start setting up your sort parameters.
        i. In the bottom left center of the screen, there will be the sort control section
            i. Select the method of collection (i.e. 5mL,96 well plate)
                a. If you decide to do plates refer to another guide. Label Plate setup.
            ii. Select the population and set the amount you want to collect
                a. On the instrument 0 mean continuous sort. If you want to have a certain number collected type it in.
            iii. Place the collection holder you are using in the sort chamber. In the designated position you had set. The collection holder fits only one way. You will need to line up the pongs to the holes under the collection holder. You know it is in place because it will not move or shake when set correctly.
            iv. Place your collection tube into the instrument and then on the software you will need to select load collection so it will be in the correct sorting position.

a. Hit start for acquiring sample, then select sort start in the sort region.
    i. Note: once you hit sort and stop the sort completely you will be forced to start a new tube
    ii. Try to limit a full stop and just hit pause when you need to switch collection tubes, add buffer, or resuspend the sample.
b. At the end of the sort click unload collection to unload and remove/replace the collection tubes.
c. Clean the sample loader, collection area, and the collection stage using ethanol to prevent salt buildup within the unit after sort is finished

b. Tips When Monitoring the Stream
    i. While the system is actively sorting. You should be monitoring the stream droplet viewer. The instrument does provide a warning a flashing color indictor (yellow/red) to the user if the stream is unsteady and will have you recalibrate when necessary.
    ii. To increase the droplet viewer image
        i. Double click and it will open into a larger window
    iii. For Troubleshooting Steps Reference Sony Troubleshooting Step 4 section a.

11) System Shutdown and Exporting Data
a. Export your PDF and FCS files off the software before performing shutdown on the software.
    i. To retrieve the FCS files
        i. Under the Experiment tab in the export section select the option export to FCS file. Checkmark all the relevant tubes or you can right click on the experiment name and select export to FCS files.
            a. Make sure you have checked the box for the option to convert the BSC to SSC in order to read these files on a third party analysis software (ex. Flowjo)
        ii. In the output folder section, you will need to direct the data to your folder. To direct it click on the three dots (…) from the Desktop->User Data file-> folders
b. Generate PDF/Sort information
    i. Under the Experiment tab select the option Custom Print after clicking on a blank space on the worksheet
        i. This will open the print dialog box found on the experiment tab on the far right. In the dialog box select the samples to be generated as a PDF. Unclick the compensation information if you did not compensation.
        ii. Hit print and direct the information to your folder. From the Desktop->user data file->folders

ii. Sort Tube Results information
    i. Only do this option if you decided not to generate a consolidated PDF file information.
    ii. Under the experiment tab in the setting and information section select show results of the tube you want to display.

c. Save the Experiment as Template
    i. Under the experiment tab in the experiment section. Select the option Save as Template or you can right click on the experiment name and select save as template.

d. Shutdown the Experiment
    i. Under the experiment tab Click Software and Hardware in the shutdown section.
        i. Follow -on screen instructions (3 step process)
            a. It will require a 15mL tube of 10% bleach at about ~12mL.
                i. Using the syringe with filter attached. Filled from with the box reagent water found on the desk. Bleach is found on the desk in a white bottle label sodium hypochlorite.
            b. Water step requires about ~12mL
                i. Using the syringe with filter attached. Filled from with the box reagent water found on the desk.
                ii. Have selected the option to keep water in the sample line. This will prevent buildup.
                iii. Remove the 15mL tube from the sample holder
            c. Click Shutdown down software and instrument
            d. Turn off the air compressor

e. Transfer Data to weTransfer
**Transferring data files:** Use a virus-free USB/thumb drive to take your files back to your lab OR use the free “wetransfer” web application to transfer your data files to a server, you can then access and download your data via a link sent to your email address. www.wetransfer.com.